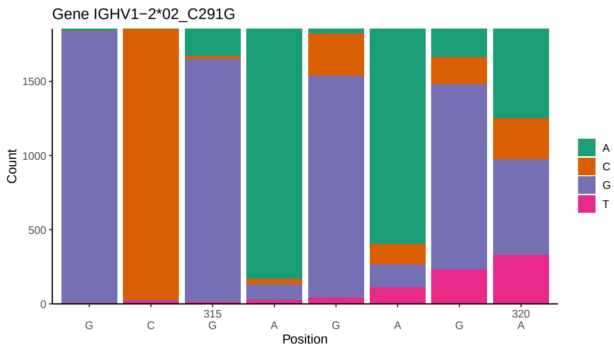
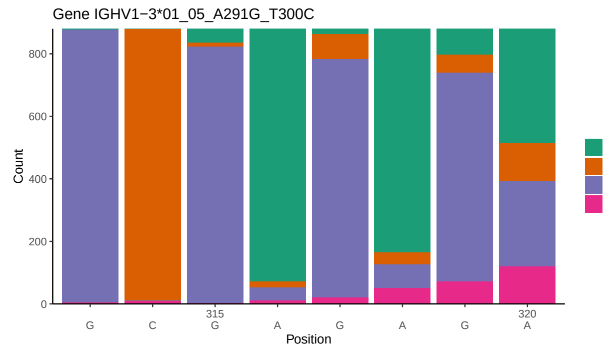
OGRDBstats Report

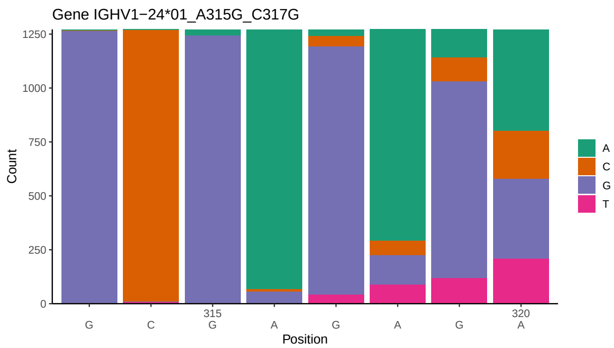
Contents

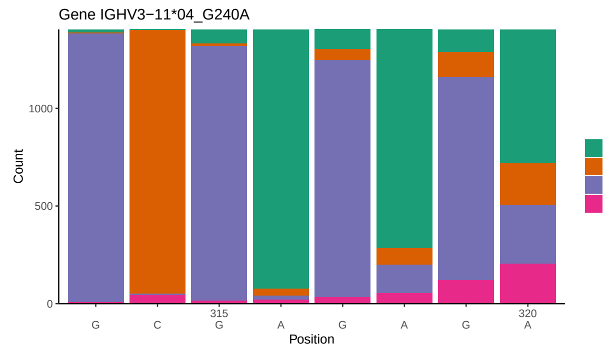
1 Novel sequence analysis

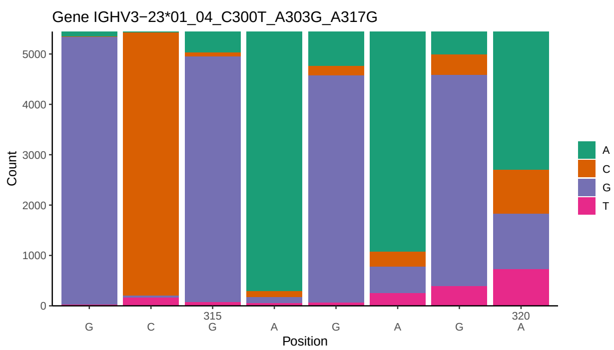
1.1 End-nucleotide composition

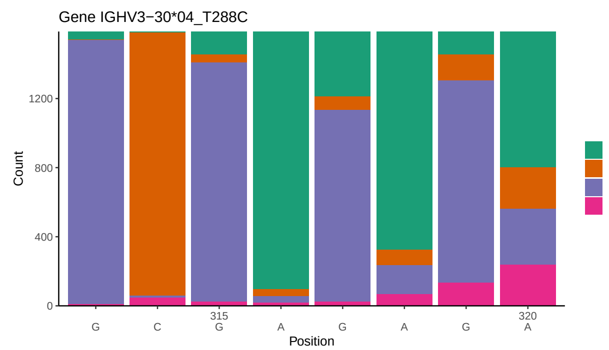


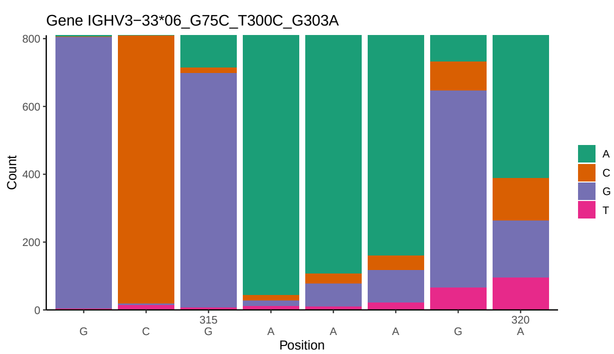


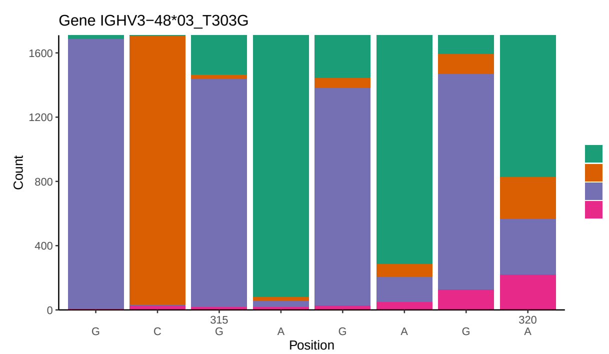


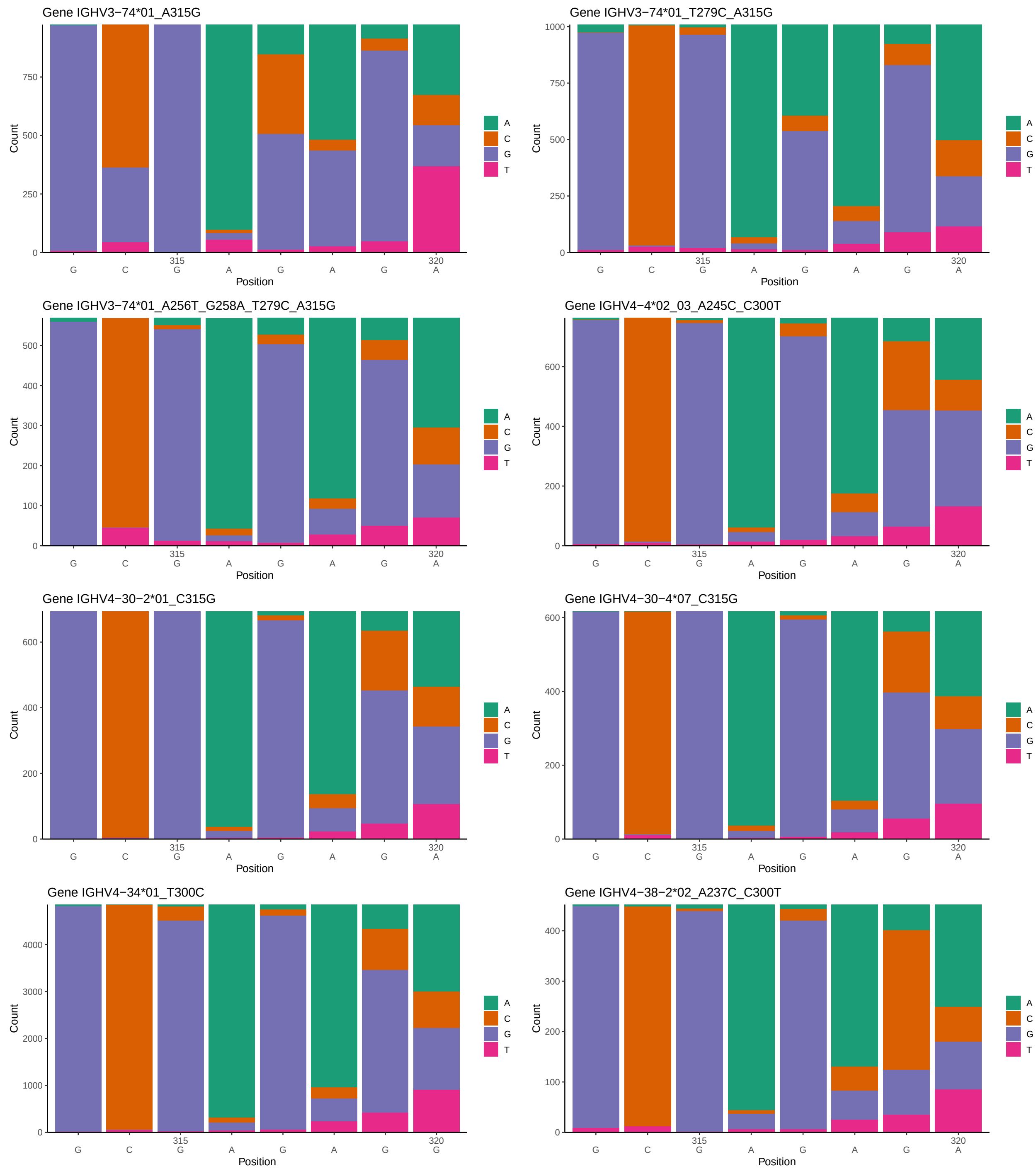


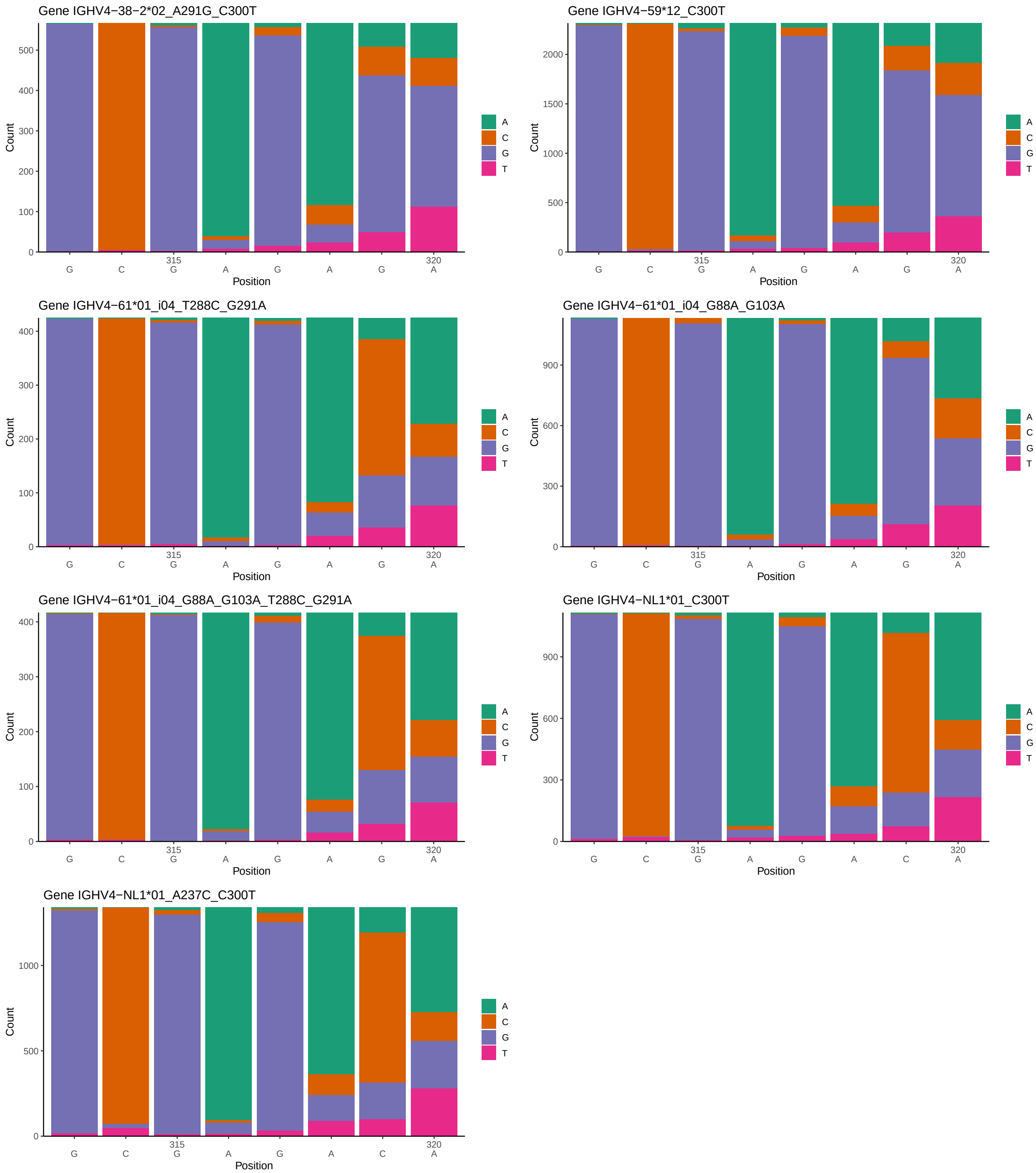




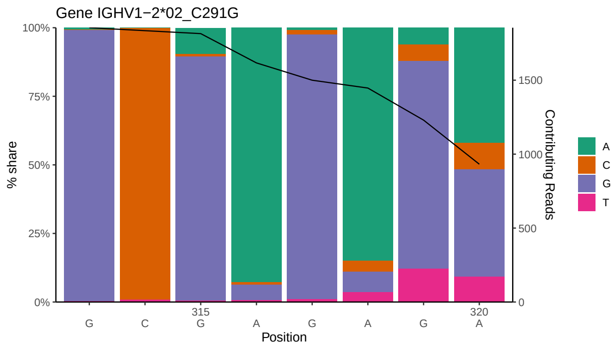


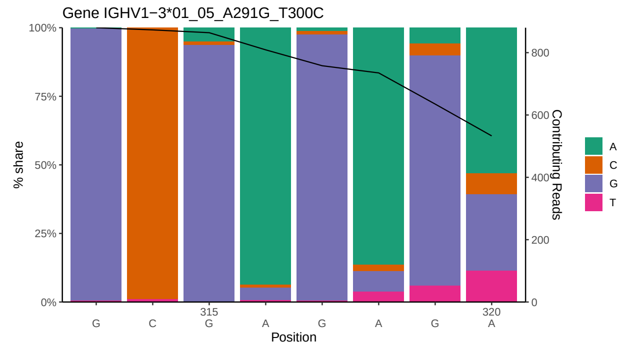


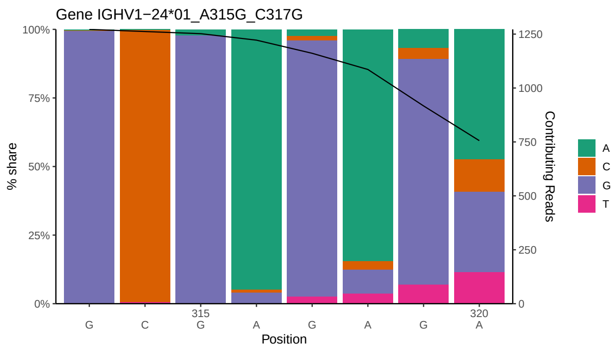

Gene IGHV3−74*01_A315G
Gene IGHV3−74*01_T279C_A315G
Gene IGHV3−74*01_A256T_G258A_T279C_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−4*07_C315G
Gene IGHV4−34*01_T300C
Gene IGHV4−38−2*02_A237C_C300T
Count
Position
A
C
G
T

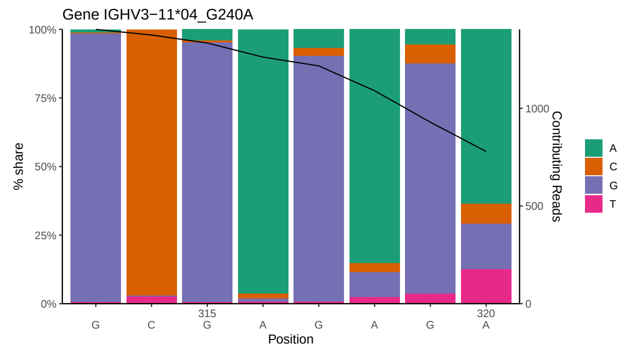

Gene IGHV4−38−2*02_A291G_C300T
Gene IGHV4−59*12_C300T
Gene IGHV4−61*01_i04_T288C_G291A
Gene IGHV4−61*01_i04_G88A_G103A
Gene IGHV4−61*01_i04_G88A_G103A_T288C_G291A
Gene IGHV4−NL1*01_C300T
Gene IGHV4−NL1*01_A237C_C300T
Count
Position
A
C
G
T

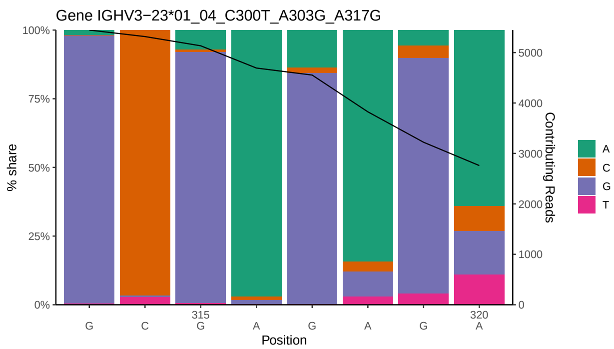
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

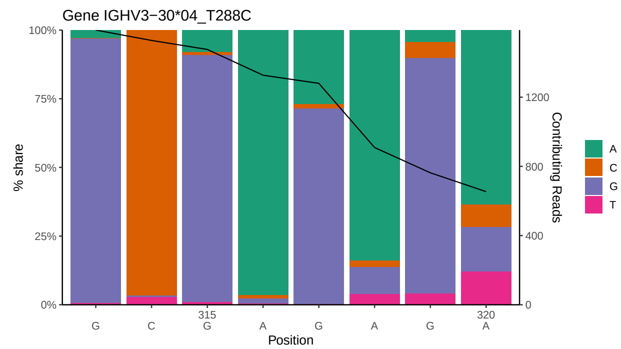


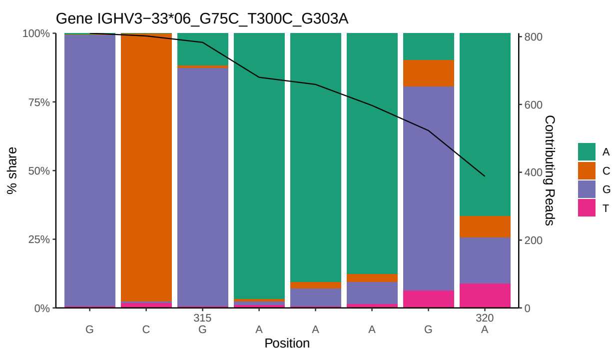


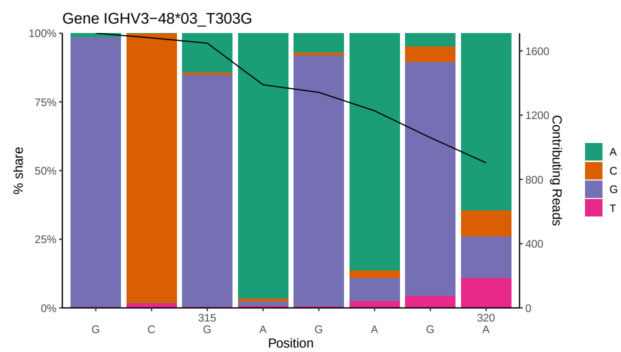


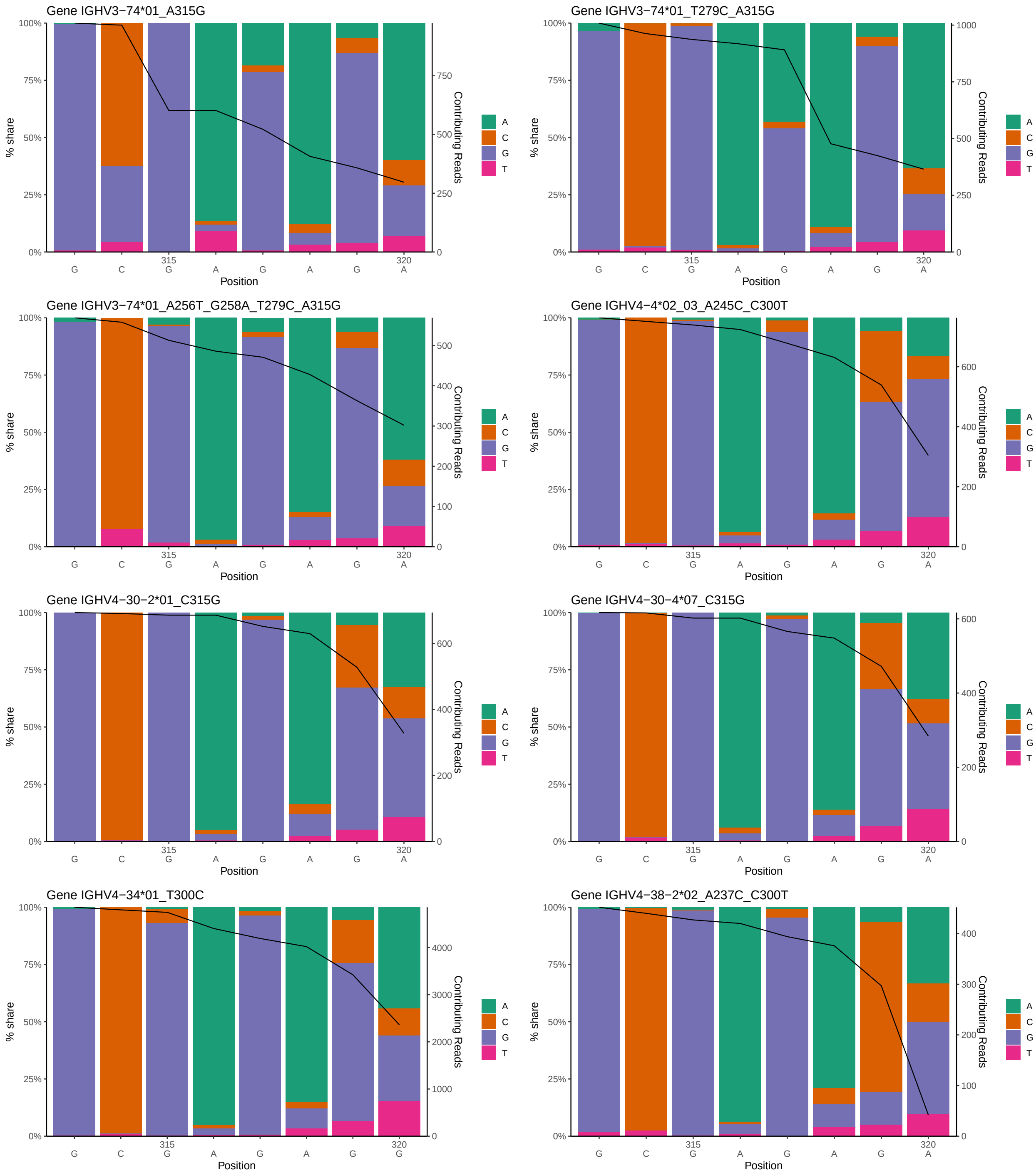




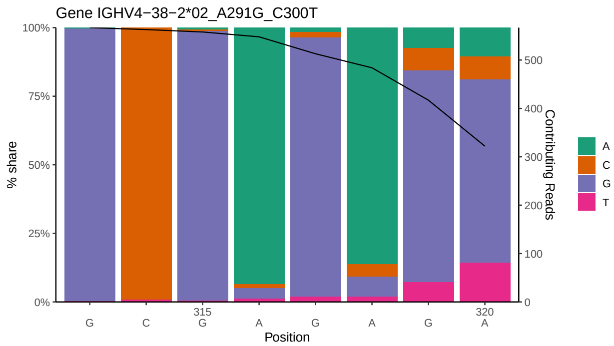


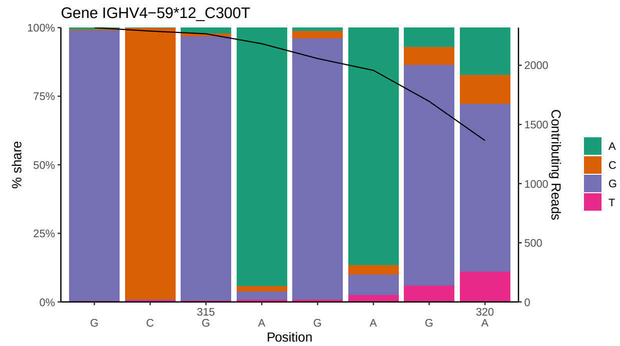


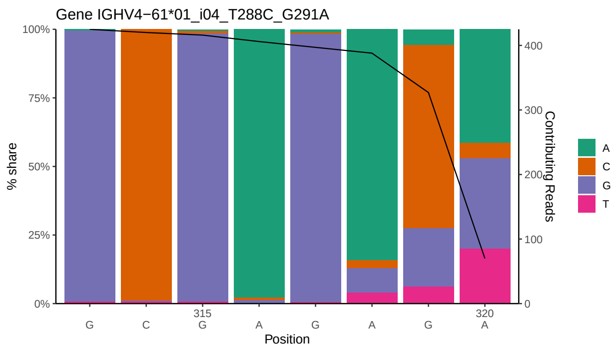


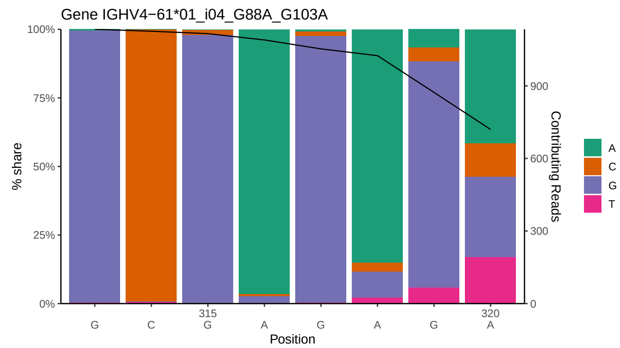


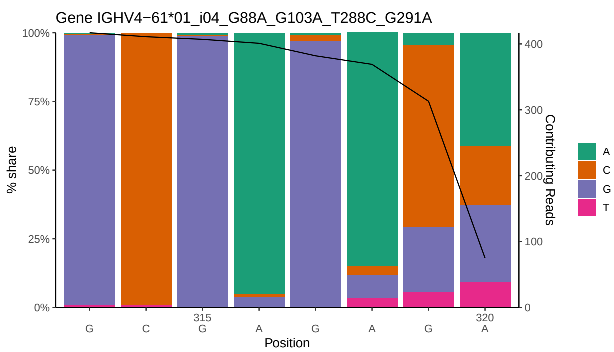
Gene IGHV3−74*01_A315G
Gene IGHV3−74*01_T279C_A315G
Gene IGHV3−74*01_A256T_G258A_T279C_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−4*07_C315G
Gene IGHV4−34*01_T300C
Gene IGHV4−38−2*02_A237C_C300T
% share
Contributing Reads
Position
A
C
G
T

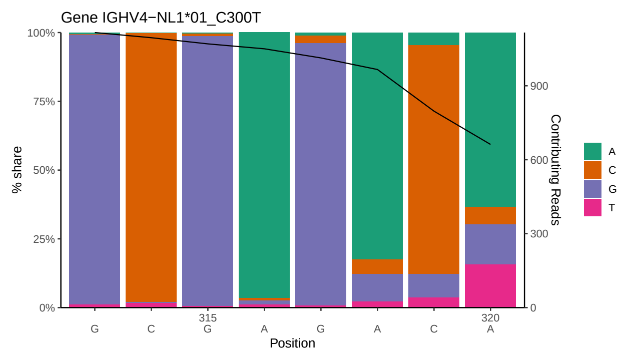

Gene IGHV4−38−2*02_A291G_C300T
% share
Contributing Reads
Position
A
C
G
T

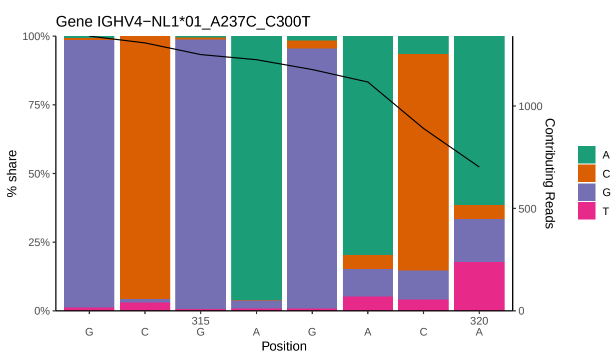

Gene IGHV4−59*12_C300T
% share
Contributing Reads
Position
A
C
G
T


Gene IGHV4−61*01_i04_T288C_G291A
% share
Contributing Reads
Position
A
C
G
T

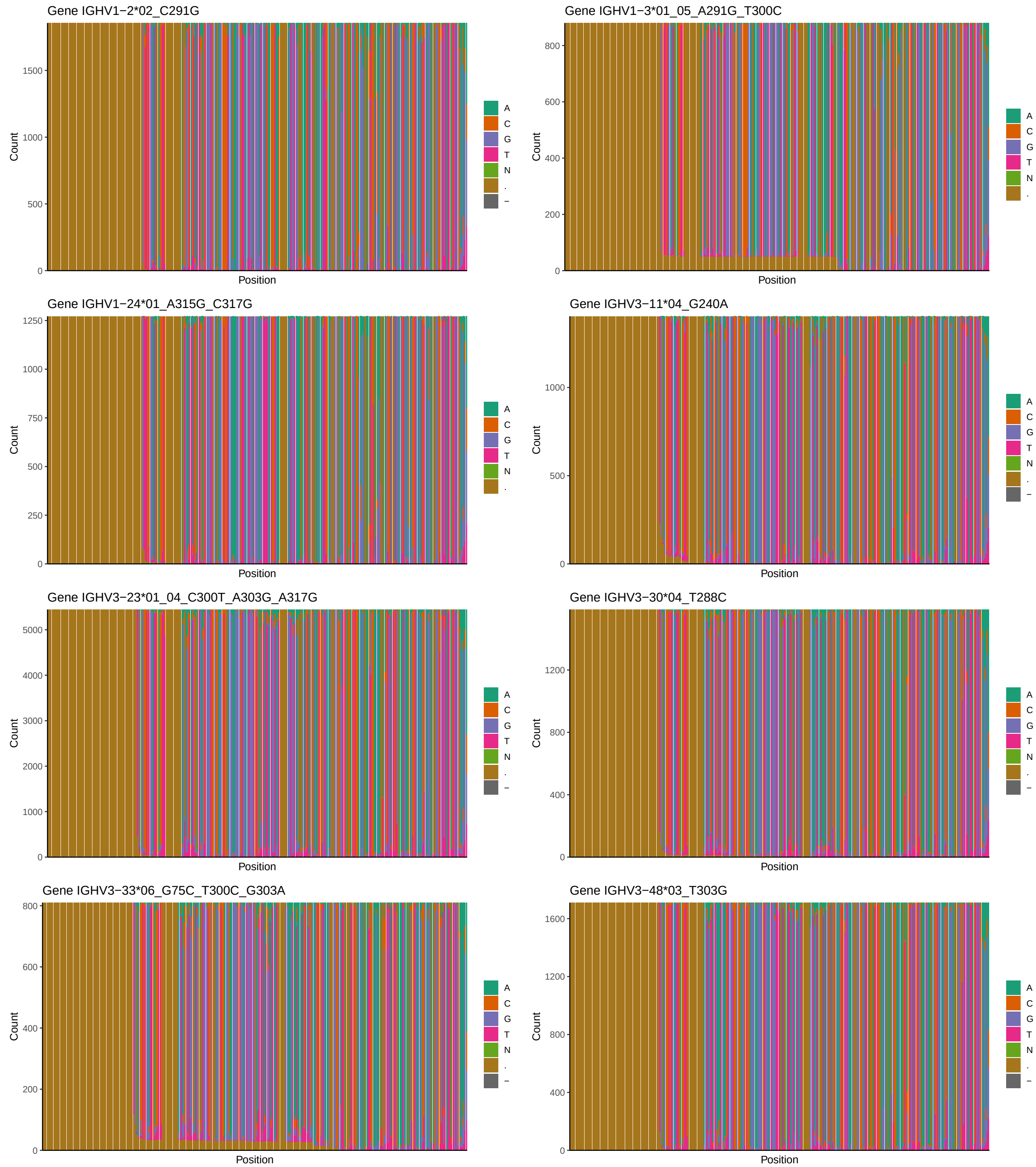

Gene IGHV4−61*01_i04_G88A_G103A
% share
Contributing Reads
Position
A
C
G
T

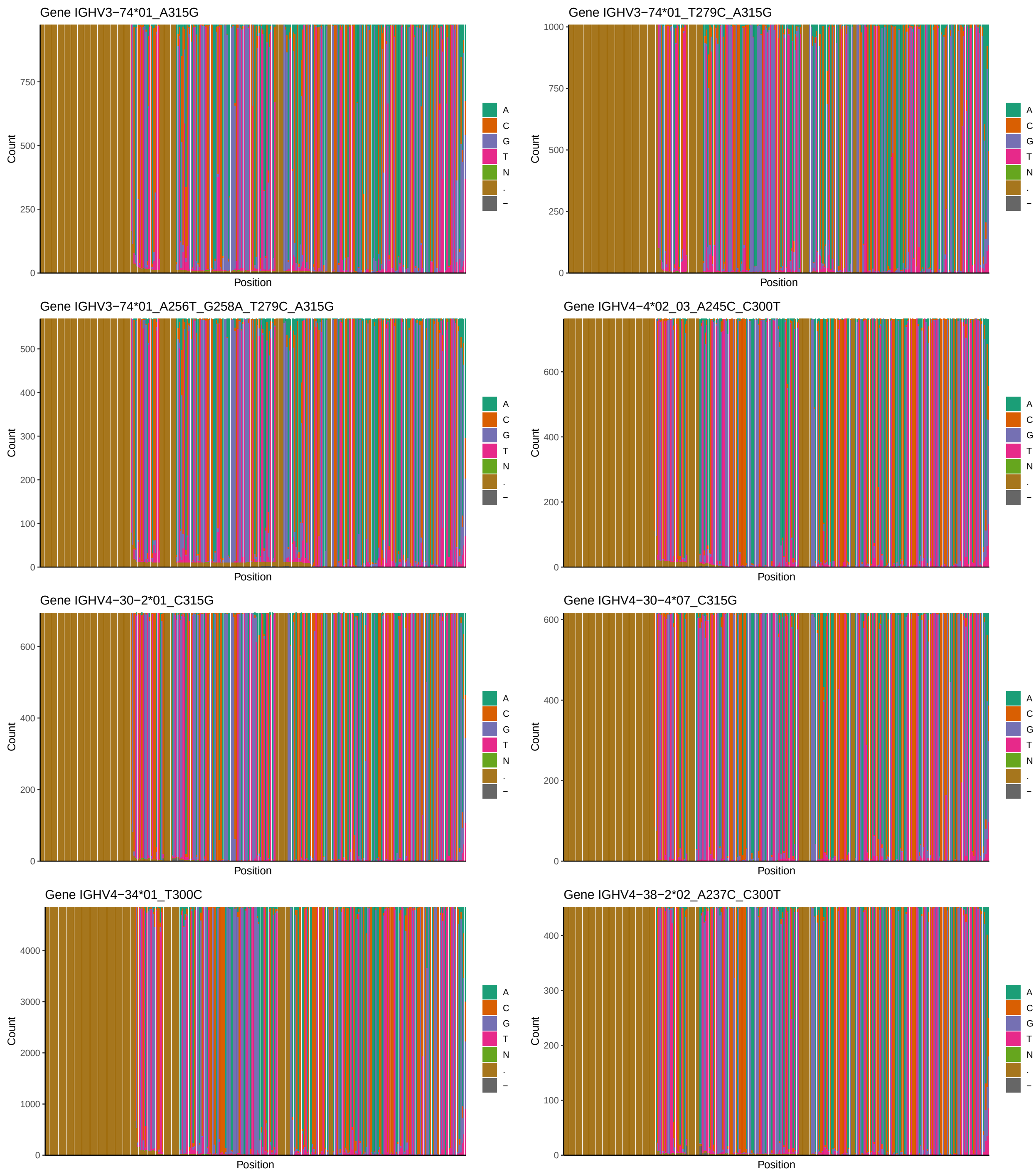

Gene IGHV4−61*01_i04_G88A_G103A_T288C_G291A
% share
Contributing Reads
Position
A
C
G
T


Gene IGHV4−NL1*01_C300T
% share
Contributing Reads
Position
A
C
G
T


Gene IGHV4−NL1*01_A237C_C300T
% share
Contributing Reads
Position
A
C
G
T

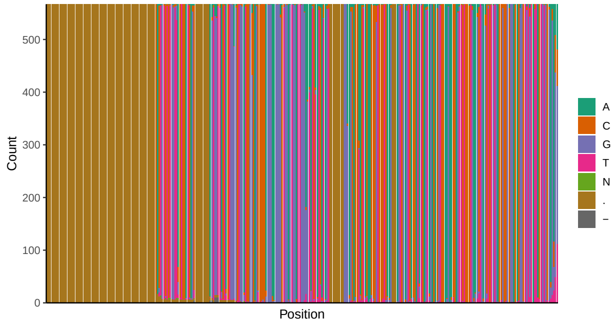
1.3 Whole-sequence composition of each assigned read



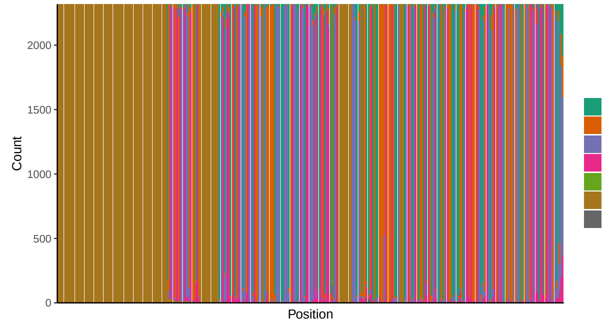


Gene IGHV3−74*01_A315G
Gene IGHV3−74*01_T279C_A315G
Gene IGHV3−74*01_A256T_G258A_T279C_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−4*07_C315G
Gene IGHV4−34*01_T300C
Gene IGHV4−38−2*02_A237C_C300T
Count
Position
A
C
G
T
N
.
−

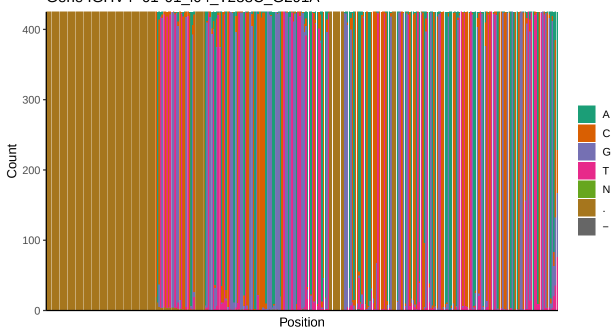
Gene IGHV4−38−2*02_A291G_C300T



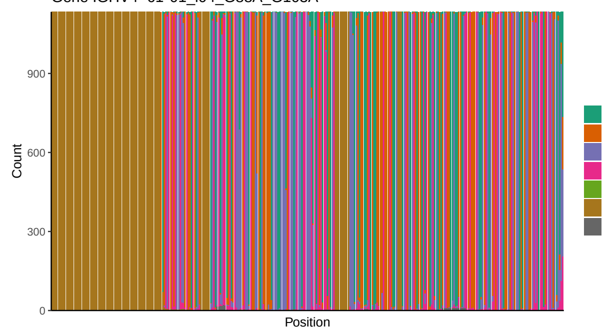
Gene IGHV4−59*12_C300T



Gene IGHV4−61*01_i04_T288C_G291A

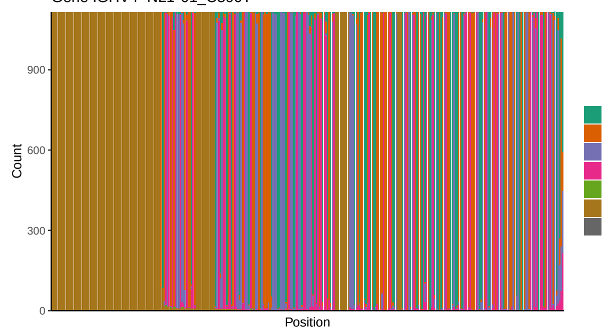


Gene IGHV4−61*01_i04_G88A_G103A

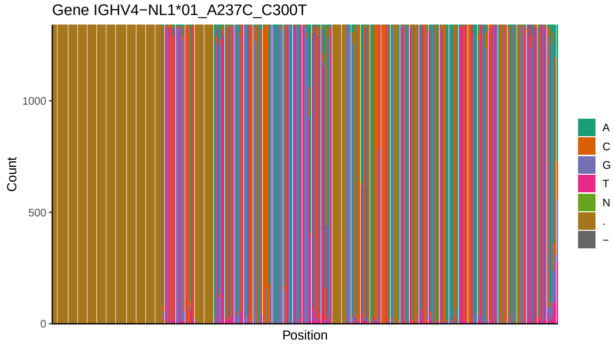


Gene IGHV4−61*01_i04_G88A_G103A_T288C_G291A

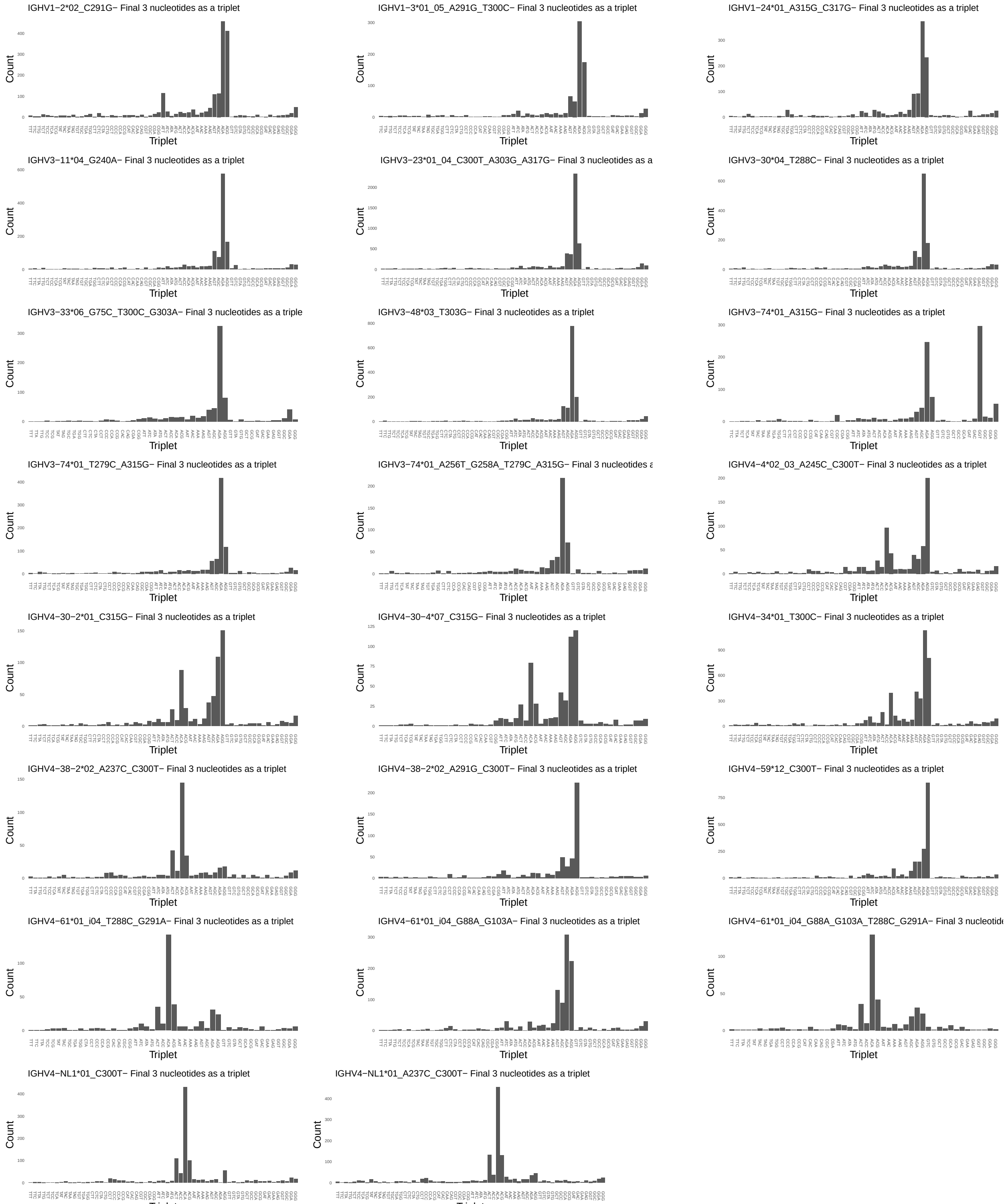
Gene IGHV4−NL1*01_C300T



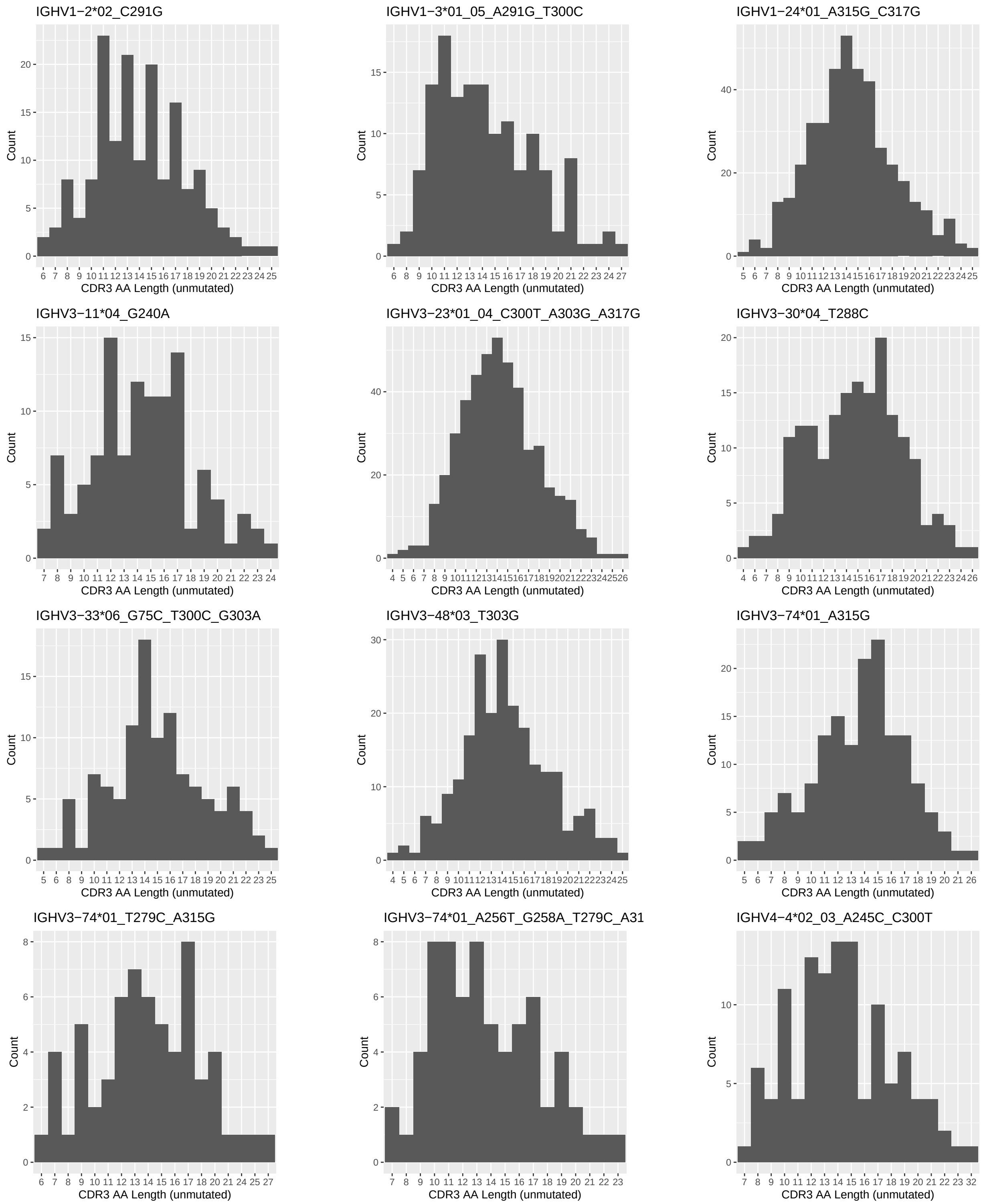
Gene IGHV4−NL1*01_A237C_C300T

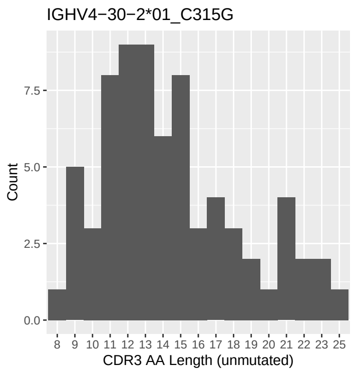
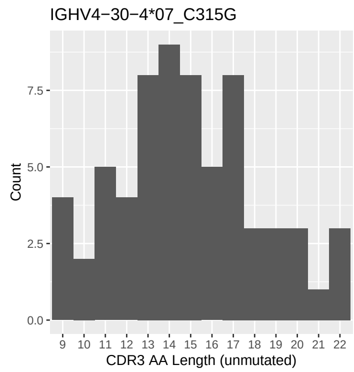
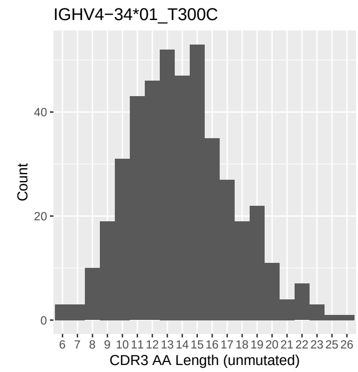
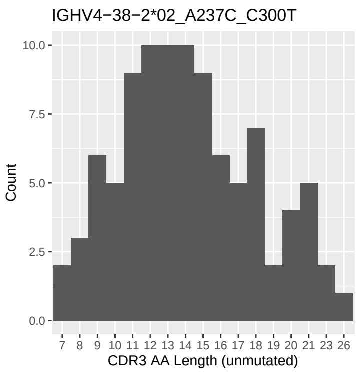
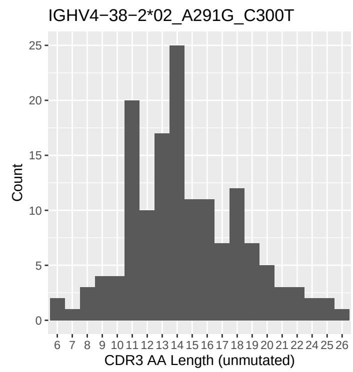
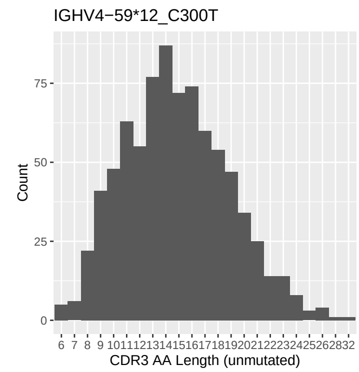
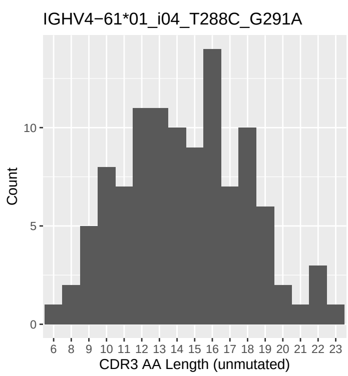
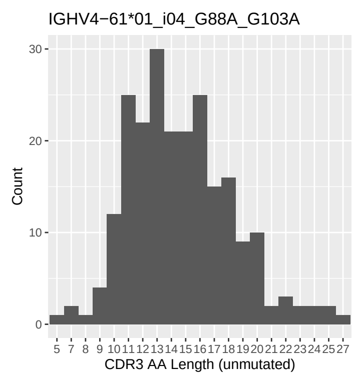
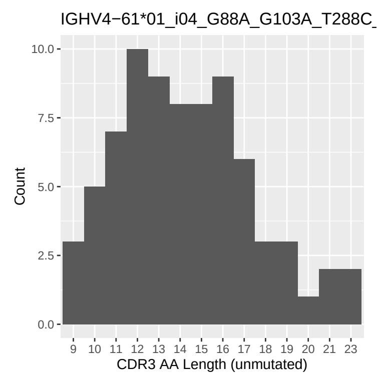
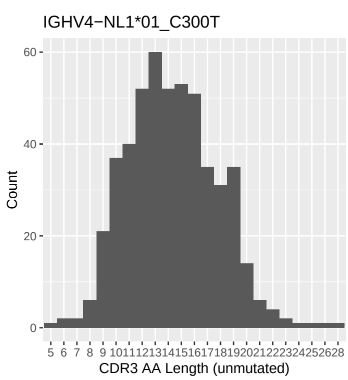
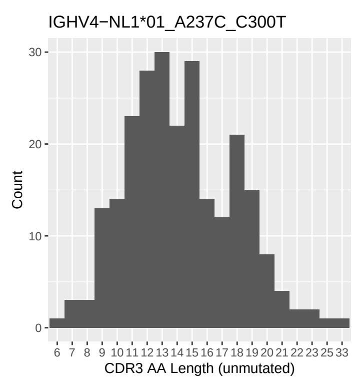


1.4 Final three nucleotides: frequency of each observed triplet

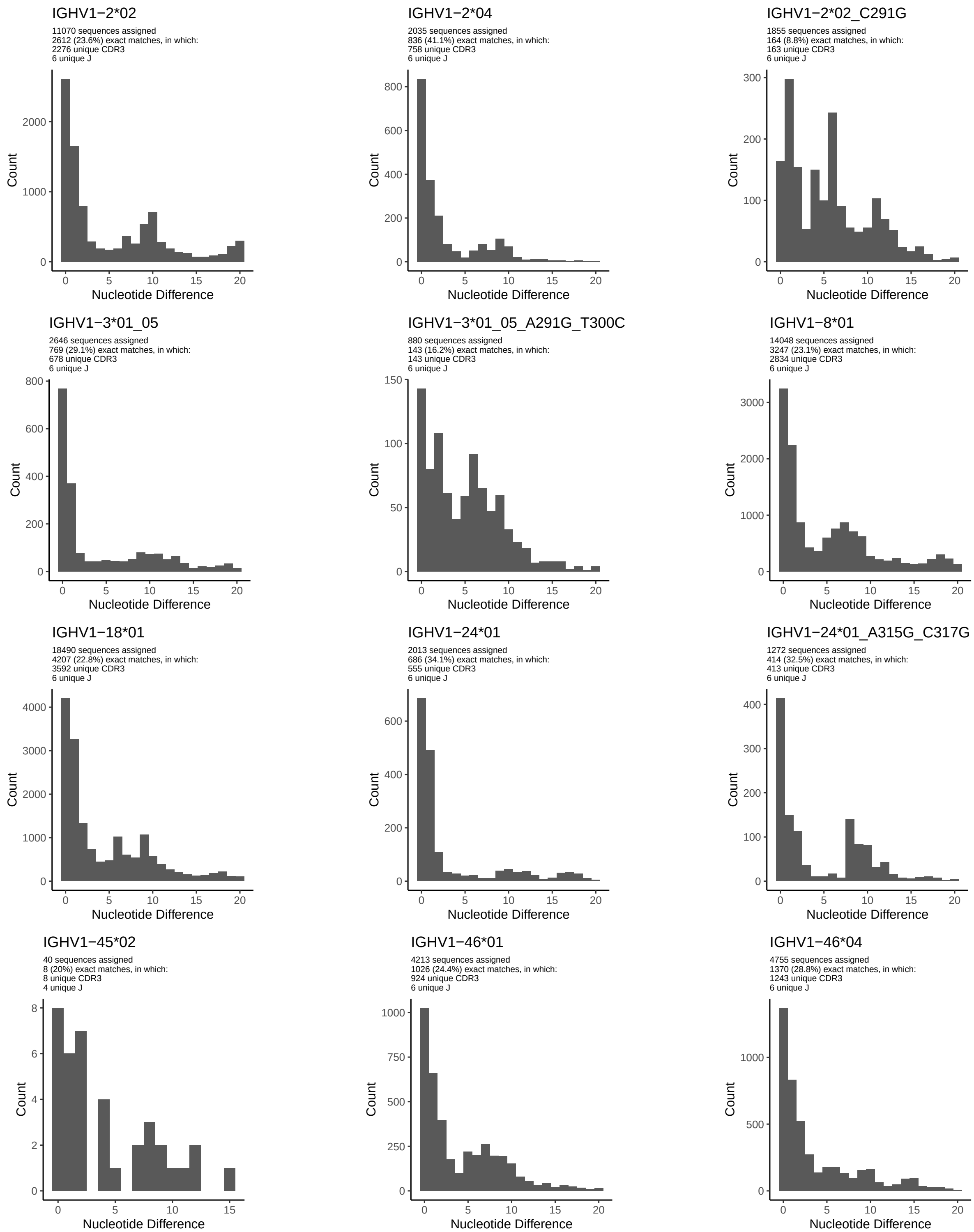


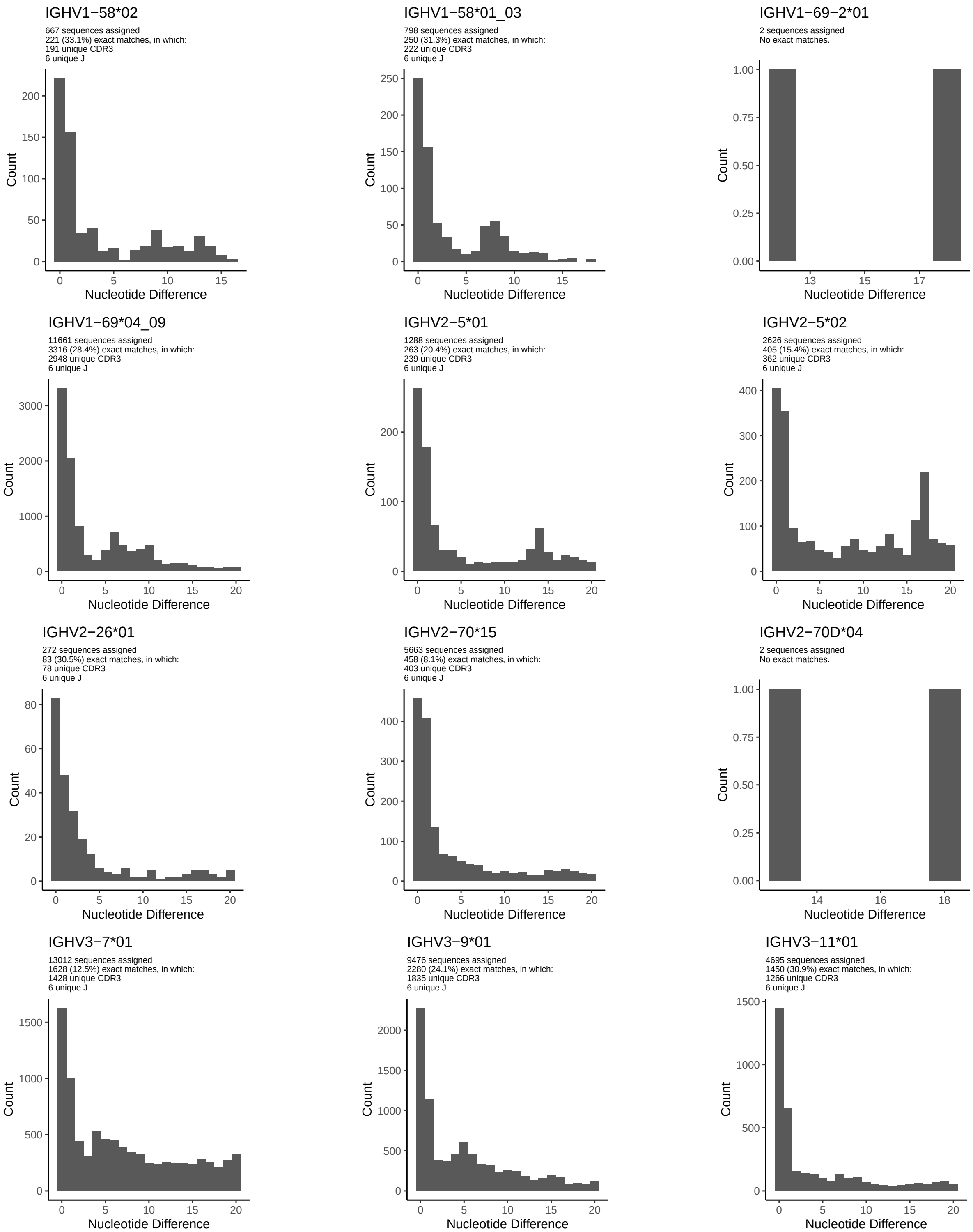
1.5 CDR3 length distribution, in assignments to novel alleles

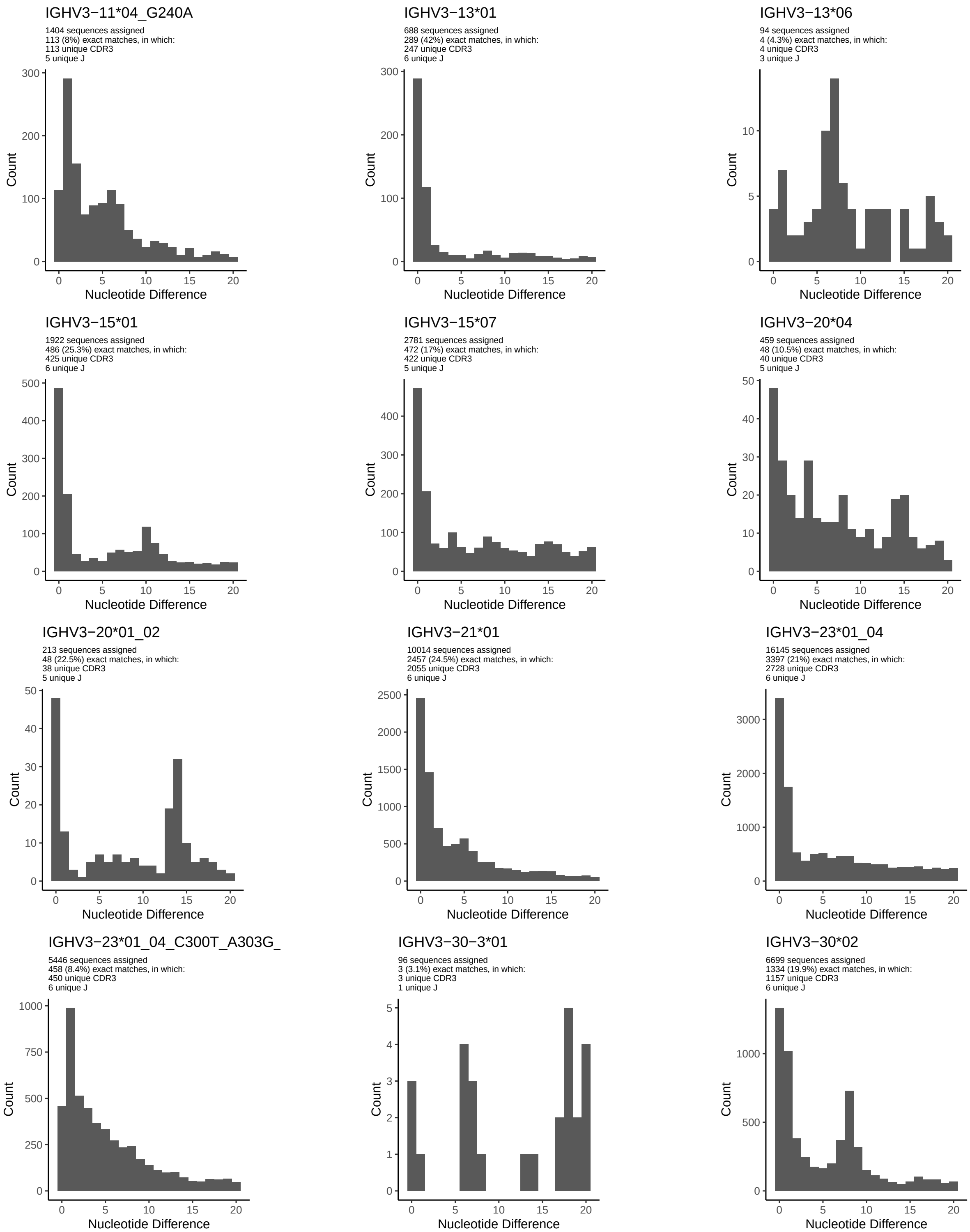


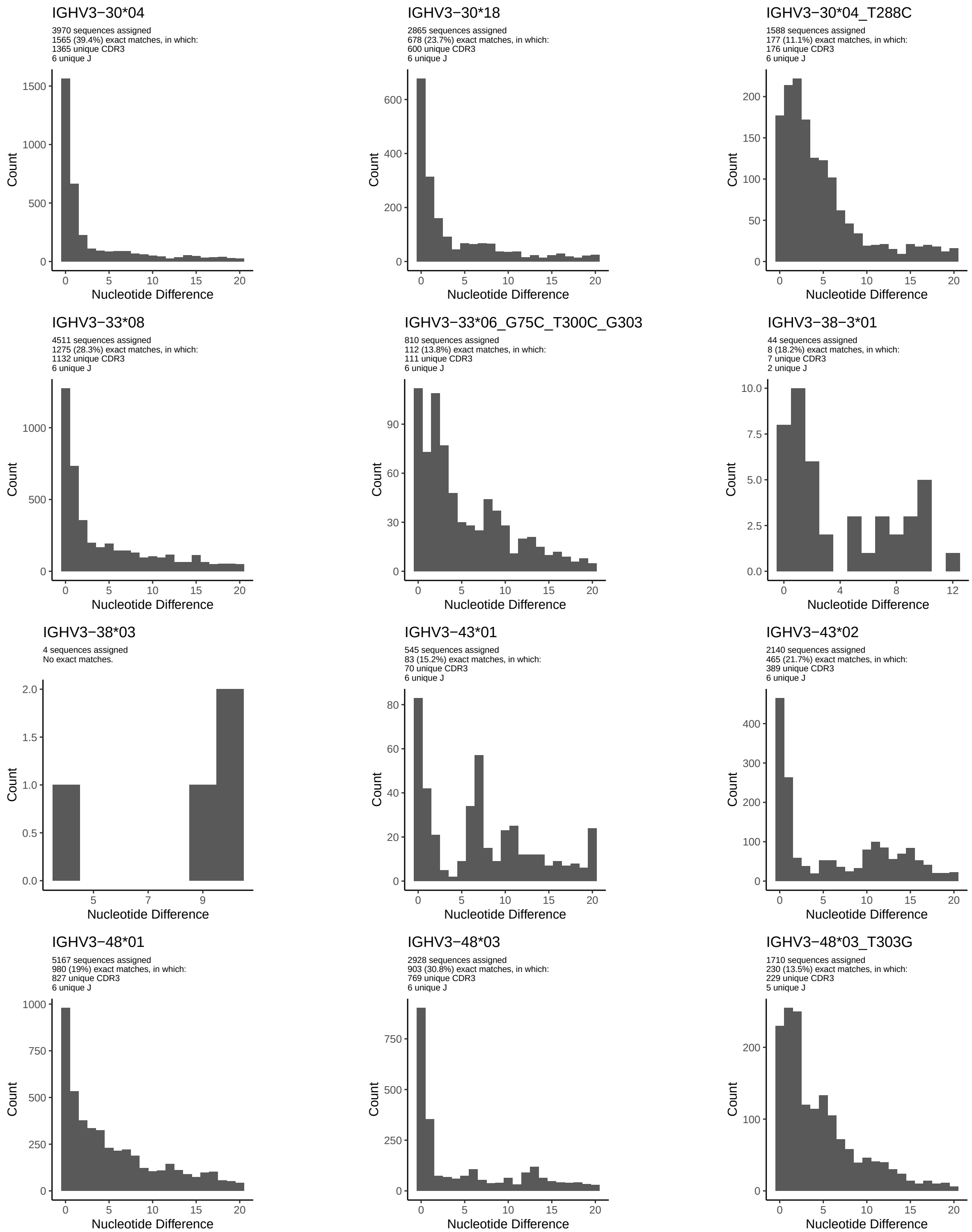

IGHV4−30−2*01_C315G
Count
CDR3 AA Length (unmutated)

IGHV4−30−4*07_C315G
Count
CDR3 AA Length (unmutated)

IGHV4−34*01_T300C
Count
CDR3 AA Length (unmutated)

IGHV4−38−2*02_A237C_C300T
Count
CDR3 AA Length (unmutated)

IGHV4−38−2*02_A291G_C300T
Count
CDR3 AA Length (unmutated)

IGHV4−59*12_C300T
Count
CDR3 AA Length (unmutated)

IGHV4−61*01_i04_T288C_G291A
Count
CDR3 AA Length (unmutated)

IGHV4−61*01_i04_G88A_G103A
Count
CDR3 AA Length (unmutated)

IGHV4−61*01_i04_G88A_G103A_T288C_
Count
CDR3 AA Length (unmutated)

IGHV4−NL1*01_C300T
Count
CDR3 AA Length (unmutated)

IGHV4−NL1*01_A237C_C300T
Count
CDR3 AA Length (unmutated)

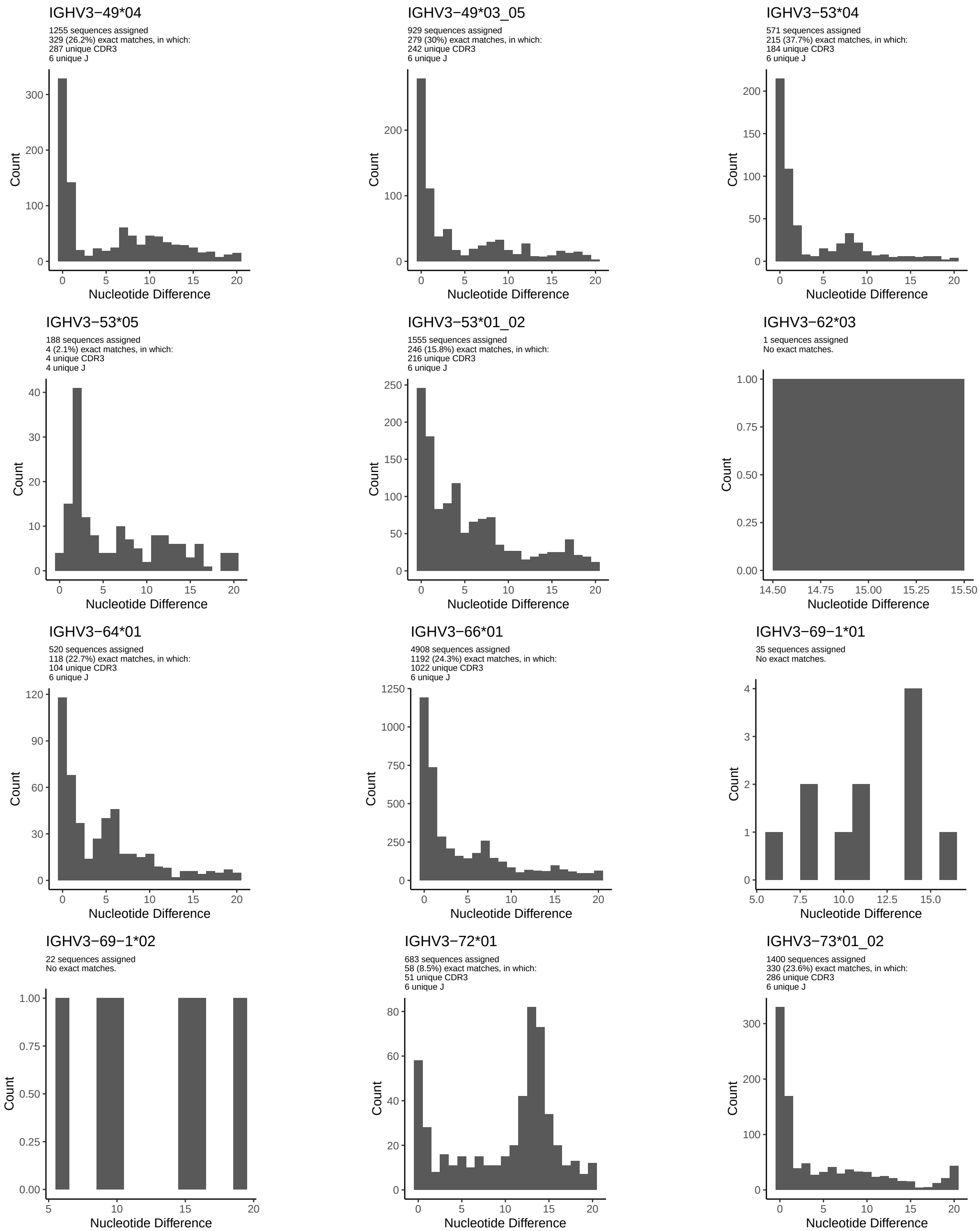
2 Variation from germline, in assignments to each allele

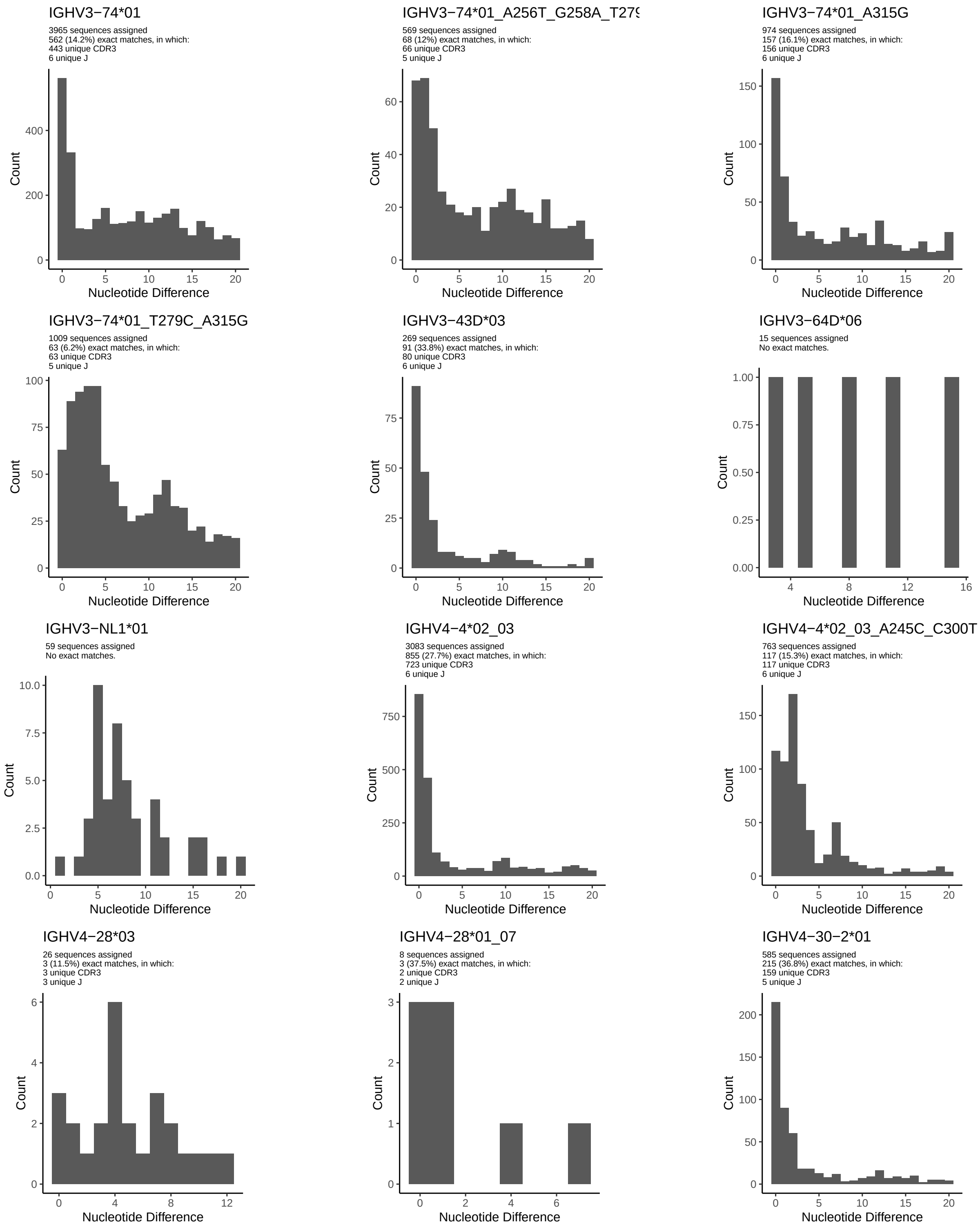


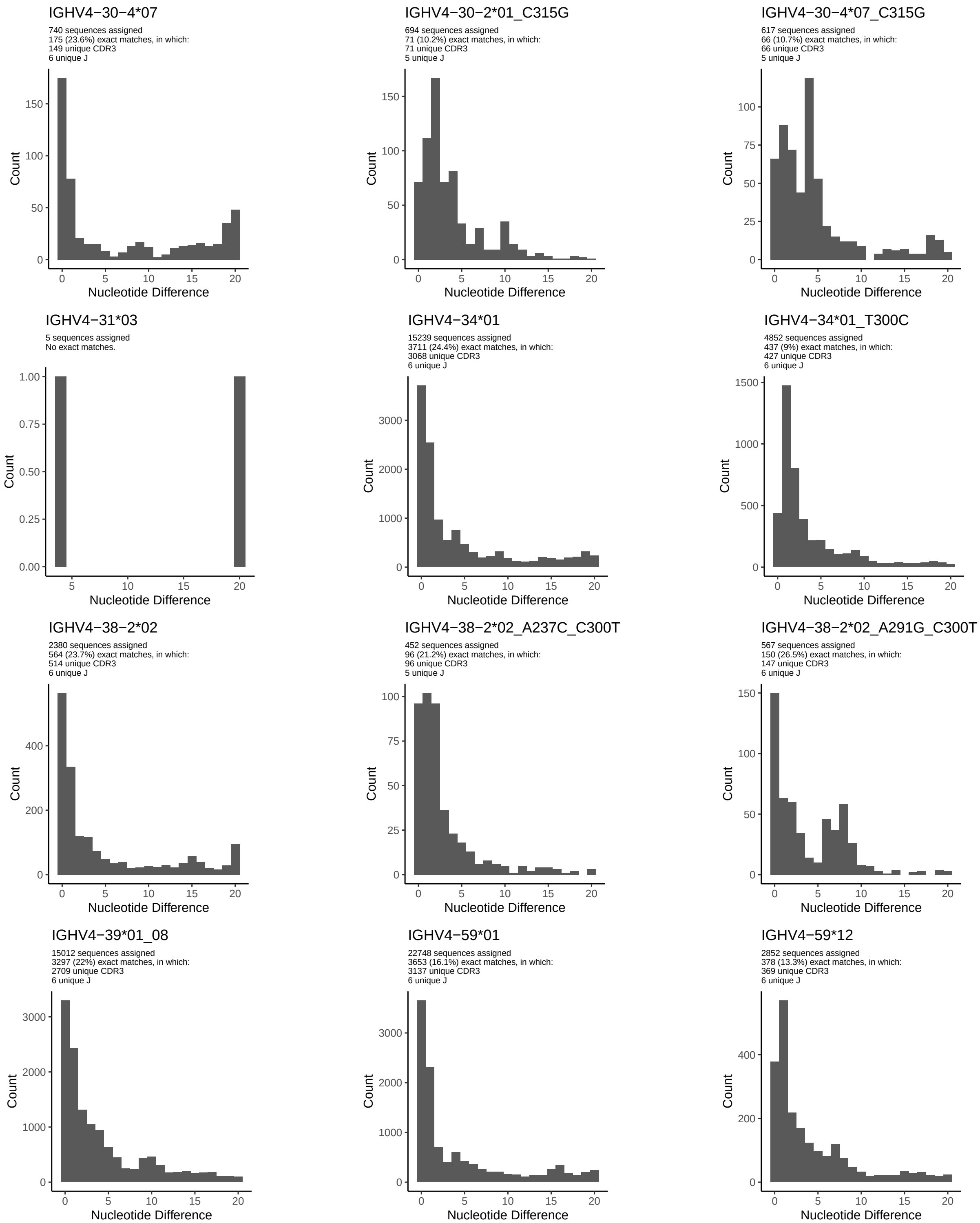

IGHV1−58*02
667 sequences assigned
221 (33.1%) exact matches, in which:
191 unique CDR3
6 unique J
IGHV1−58*01_03
798 sequences assigned
250 (31.3%) exact matches, in which:
222 unique CDR3
6 unique J
IGHV1−69−2*01
2 sequences assigned
No exact matches.
IGHV1−69*04_09
11661 sequences assigned
3316 (28.4%) exact matches, in which:
2948 unique CDR3
6 unique J
IGHV2−5*01
1288 sequences assigned
263 (20.4%) exact matches, in which:
239 unique CDR3
6 unique J
IGHV2−5*02
2626 sequences assigned
405 (15.4%) exact matches, in which:
362 unique CDR3
6 unique J
IGHV2−26*01
272 sequences assigned
83 (30.5%) exact matches, in which:
78 unique CDR3
6 unique J
IGHV2−70*15
5663 sequences assigned
458 (8.1%) exact matches, in which:
403 unique CDR3
6 unique J
IGHV2−70D*04
2 sequences assigned
No exact matches.
IGHV3−7*01
13012 sequences assigned
1628 (12.5%) exact matches, in which:
1428 unique CDR3
6 unique J
IGHV3−9*01
9476 sequences assigned
2280 (24.1%) exact matches, in which:
1835 unique CDR3
6 unique J
IGHV3−11*01
4695 sequences assigned
1450 (30.9%) exact matches, in which:
1266 unique CDR3
6 unique J
Count
Nucleotide Difference

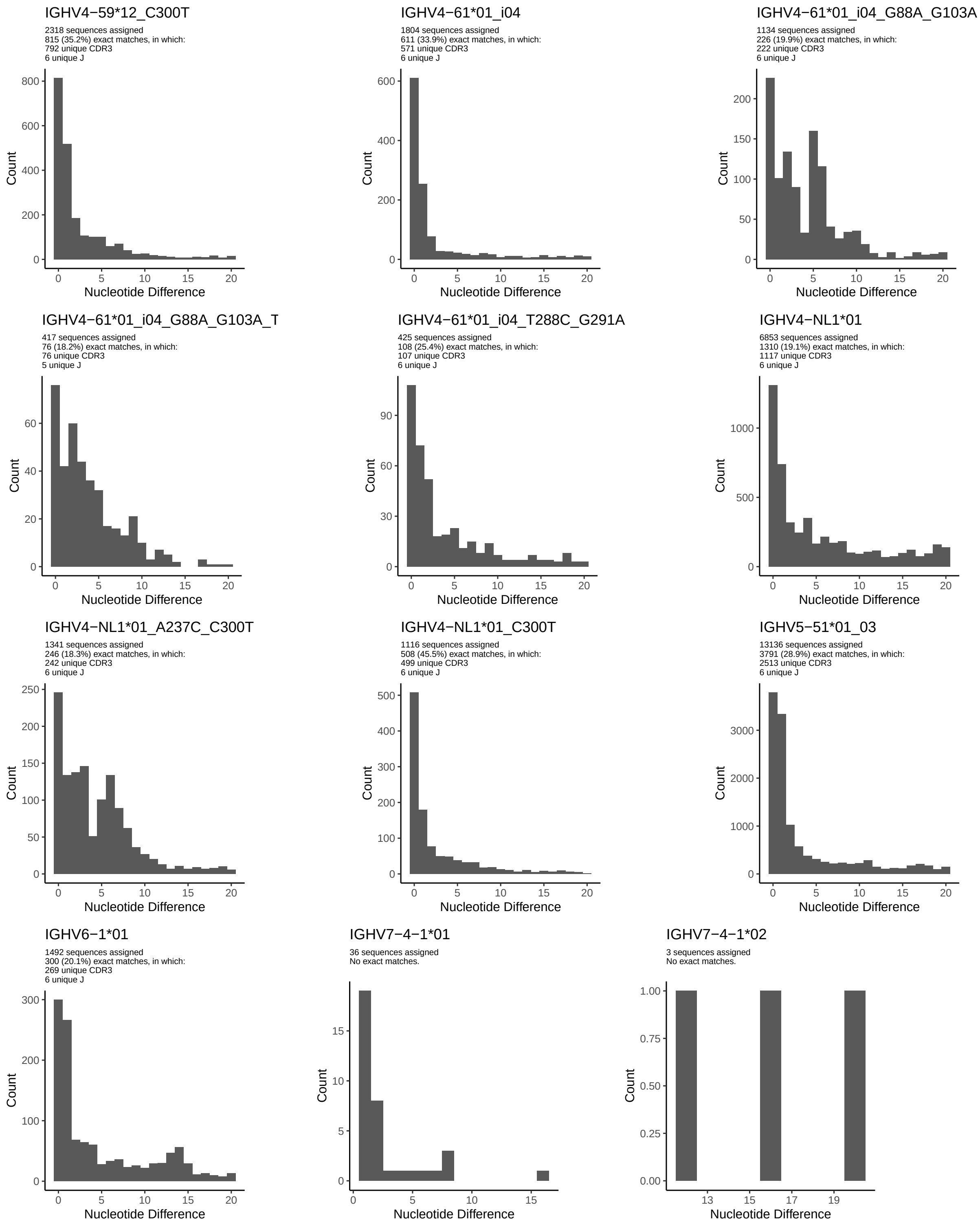

IGHV3−11*04_G240A
1404 sequences assigned
113 (8%) exact matches, in which:
113 unique CDR3
5 unique J
IGHV3−13*01
688 sequences assigned
289 (42%) exact matches, in which:
247 unique CDR3
6 unique J
IGHV3−13*06
94 sequences assigned
4 (4.3%) exact matches, in which:
4 unique CDR3
3 unique J
IGHV3−15*01
1922 sequences assigned
486 (25.3%) exact matches, in which:
425 unique CDR3
6 unique J
IGHV3−15*07
2781 sequences assigned
472 (17%) exact matches, in which:
422 unique CDR3
5 unique J
IGHV3−20*04
459 sequences assigned
48 (10.5%) exact matches, in which:
40 unique CDR3
5 unique J
IGHV3−20*01_02
213 sequences assigned
48 (22.5%) exact matches, in which:
38 unique CDR3
5 unique J
IGHV3−21*01
10014 sequences assigned
2457 (24.5%) exact matches, in which:
2055 unique CDR3
6 unique J
IGHV3−23*01_04
16145 sequences assigned
3397 (21%) exact matches, in which:
2728 unique CDR3
6 unique J
IGHV3−23*01_04_C300T_A303G_
5446 sequences assigned
458 (8.4%) exact matches, in which:
450 unique CDR3
6 unique J
IGHV3−30−3*01
96 sequences assigned
3 (3.1%) exact matches, in which:
3 unique CDR3
1 unique J
IGHV3−30*02
6699 sequences assigned
1334 (19.9%) exact matches, in which:
1157 unique CDR3
6 unique J


IGHV3−30*04
3970 sequences assigned
1565 (39.4%) exact matches, in which:
1365 unique CDR3
6 unique J
IGHV3−30*18
2865 sequences assigned
678 (23.7%) exact matches, in which:
600 unique CDR3
6 unique J
IGHV3−30*04_T288C
1588 sequences assigned
177 (11.1%) exact matches, in which:
176 unique CDR3
6 unique J
IGHV3−33*08
4511 sequences assigned
1275 (28.3%) exact matches, in which:
1132 unique CDR3
6 unique J
IGHV3−33*06_G75C_T300C_G303
810 sequences assigned
112 (13.8%) exact matches, in which:
111 unique CDR3
6 unique J
IGHV3−38−3*01
44 sequences assigned
8 (18.2%) exact matches, in which:
7 unique CDR3
2 unique J
IGHV3−38*03
4 sequences assigned
No exact matches.
IGHV3−43*01
545 sequences assigned
83 (15.2%) exact matches, in which:
70 unique CDR3
6 unique J
IGHV3−43*02
2140 sequences assigned
465 (21.7%) exact matches, in which:
389 unique CDR3
6 unique J
IGHV3−48*01
5167 sequences assigned
980 (19%) exact matches, in which:
827 unique CDR3
6 unique J
IGHV3−48*03
2928 sequences assigned
903 (30.8%) exact matches, in which:
769 unique CDR3
6 unique J
IGHV3−48*03_T303G
1710 sequences assigned
230 (13.5%) exact matches, in which:
229 unique CDR3
5 unique J
Count
Nucleotide Difference

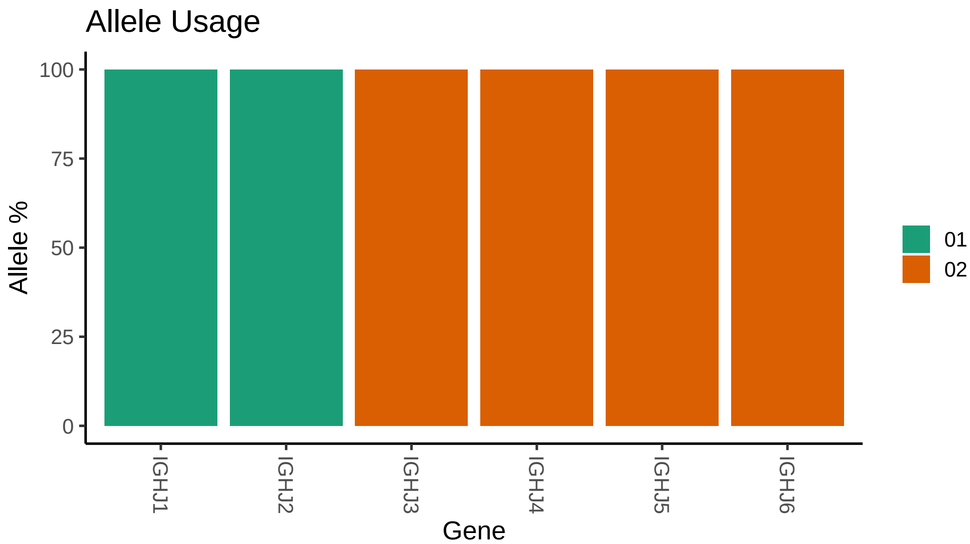

IGHV3−49*04
1255 sequences assigned
329 (26.2%) exact matches, in which:
287 unique CDR3
6 unique J
IGHV3−49*03_05
929 sequences assigned
279 (30%) exact matches, in which:
242 unique CDR3
6 unique J
IGHV3−53*04
571 sequences assigned
215 (37.7%) exact matches, in which:
184 unique CDR3
6 unique J
IGHV3−53*05
188 sequences assigned
4 (2.1%) exact matches, in which:
4 unique CDR3
4 unique J
IGHV3−53*01_02
1555 sequences assigned
246 (15.8%) exact matches, in which:
216 unique CDR3
6 unique J
IGHV3−62*03
1 sequences assigned
No exact matches.
IGHV3−64*01
520 sequences assigned
118 (22.7%) exact matches, in which:
104 unique CDR3
6 unique J
IGHV3−66*01
4908 sequences assigned
1192 (24.3%) exact matches, in which:
1022 unique CDR3
6 unique J
IGHV3−69−1*01
35 sequences assigned
No exact matches.
IGHV3−69−1*02
22 sequences assigned
No exact matches.
IGHV3−72*01
683 sequences assigned
58 (8.5%) exact matches, in which:
51 unique CDR3
6 unique J
IGHV3−73*01_02
1400 sequences assigned
330 (23.6%) exact matches, in which:
286 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV3−74*01
3965 sequences assigned
562 (14.2%) exact matches, in which:
443 unique CDR3
6 unique J
IGHV3−74*01_A256T_G258A_T27
569 sequences assigned
68 (12%) exact matches, in which:
66 unique CDR3
5 unique J
IGHV3−74*01_A315G
974 sequences assigned
157 (16.1%) exact matches, in which:
156 unique CDR3
6 unique J
IGHV3−74*01_T279C_A315G
1009 sequences assigned
63 (6.2%) exact matches, in which:
63 unique CDR3
5 unique J
IGHV3−43D*03
269 sequences assigned
91 (33.8%) exact matches, in which:
80 unique CDR3
6 unique J
IGHV3−64D*06
15 sequences assigned
No exact matches.
IGHV3−NL1*01
59 sequences assigned
No exact matches.
IGHV4−4*02_03
3083 sequences assigned
855 (27.7%) exact matches, in which:
723 unique CDR3
6 unique J
IGHV4−4*02_03_A245C_C300T
763 sequences assigned
117 (15.3%) exact matches, in which:
117 unique CDR3
6 unique J
IGHV4−28*03
26 sequences assigned
3 (11.5%) exact matches, in which:
3 unique CDR3
3 unique J
IGHV4−28*01_07
8 sequences assigned
3 (37.5%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV4−30−2*01
585 sequences assigned
215 (36.8%) exact matches, in which:
159 unique CDR3
5 unique J
Count
Nucleotide Difference


IGHV4−30−4*07
740 sequences assigned
175 (23.6%) exact matches, in which:
149 unique CDR3
6 unique J
IGHV4−30−2*01_C315G
694 sequences assigned
71 (10.2%) exact matches, in which:
71 unique CDR3
5 unique J
IGHV4−30−4*07_C315G
617 sequences assigned
66 (10.7%) exact matches, in which:
66 unique CDR3
5 unique J
IGHV4−31*03
5 sequences assigned
No exact matches.
IGHV4−34*01
15239 sequences assigned
3711 (24.4%) exact matches, in which:
3068 unique CDR3
6 unique J
IGHV4−34*01_T300C
4852 sequences assigned
437 (9%) exact matches, in which:
427 unique CDR3
6 unique J
IGHV4−38−2*02
2380 sequences assigned
564 (23.7%) exact matches, in which:
514 unique CDR3
6 unique J
IGHV4−38−2*02_A237C_C300T
452 sequences assigned
96 (21.2%) exact matches, in which:
96 unique CDR3
5 unique J
IGHV4−38−2*02_A291G_C300T
567 sequences assigned
150 (26.5%) exact matches, in which:
147 unique CDR3
6 unique J
IGHV4−39*01_08
15012 sequences assigned
3297 (22%) exact matches, in which:
2709 unique CDR3
6 unique J
IGHV4−59*01
22748 sequences assigned
3653 (16.1%) exact matches, in which:
3137 unique CDR3
6 unique J
IGHV4−59*12
2852 sequences assigned
378 (13.3%) exact matches, in which:
369 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−59*12_C300T
2318 sequences assigned
815 (35.2%) exact matches, in which:
792 unique CDR3
6 unique J
IGHV4−61*01_i04
1804 sequences assigned
611 (33.9%) exact matches, in which:
571 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_G103A
1134 sequences assigned
226 (19.9%) exact matches, in which:
222 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_G103A_T
417 sequences assigned
76 (18.2%) exact matches, in which:
76 unique CDR3
5 unique J
IGHV4−61*01_i04_T288C_G291A
425 sequences assigned
108 (25.4%) exact matches, in which:
107 unique CDR3
6 unique J
IGHV4−NL1*01
6853 sequences assigned
1310 (19.1%) exact matches, in which:
1117 unique CDR3
6 unique J
IGHV4−NL1*01_A237C_C300T
1341 sequences assigned
246 (18.3%) exact matches, in which:
242 unique CDR3
6 unique J
IGHV4−NL1*01_C300T
1116 sequences assigned
508 (45.5%) exact matches, in which:
499 unique CDR3
6 unique J
IGHV5−51*01_03
13136 sequences assigned
3791 (28.9%) exact matches, in which:
2513 unique CDR3
6 unique J
IGHV6−1*01
1492 sequences assigned
300 (20.1%) exact matches, in which:
269 unique CDR3
6 unique J
IGHV7−4−1*01
36 sequences assigned
No exact matches.
IGHV7−4−1*02
3 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /work/jenkins/PRJNA491287/M64 054/M64 054/M64 054/M64 054/14/f2c8ad38ba3a8095c7ef20
##
## Germline reference file: /work/jenkins/PRJNA491287/M64 054/M64 054/M64 054/M64 054/14/f2c8ad38ba3a80
##
## Novel allele file: /work/jenkins/PRJNA491287/M64 054/M64 054/M64 054/M64 054/14/f2c8ad38ba3a8095c7ef
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 2 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 2 02_C291G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 11 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 11 04_G240A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 04_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 08: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_G75C_T300C_G303A: replacing with gap
```

```
##
##
## Unexpected N in reference sequence IGHV3 48 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 03_T303G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_T279C_A315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A256T_G258A_T279C_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 4 07: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 07_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 38 2 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 02_A237C_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 02_A291G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 61 01_i04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_T288C_G291A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_G103A: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_G103A_T288C_G291A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_A237C_C300T: replacing with gap
```